



DiaTool-DPO

Multi-Turn Direct Preference Optimization
for Tool-Augmented LLMs



Automatic paired trajectory construction for dialogue flow control
without human labeling



SFT Alone Fails to Teach Slot-Filling and Tool Rejection

TA-LLMs must decide: ask follow-up questions, make tool calls, or reject inappropriate requests. SFT only teaches what TO do — not what NOT to do.



Failure Mode 1

Missing Follow-Up Questions

User provides incomplete info
(e.g., "Book a flight" with no
destination or date specified).

SFT model skips slot-filling and
proceeds to hallucinate arguments.

Hallucination



Failure Mode 2

Hallucinated Tool Invocations

Model invokes tools with
fabricated or incorrect argument
values instead of asking user.

Results in wrong API calls and
downstream errors in production.

Slot Hallucination



Failure Mode 3

Failed Tool Rejection

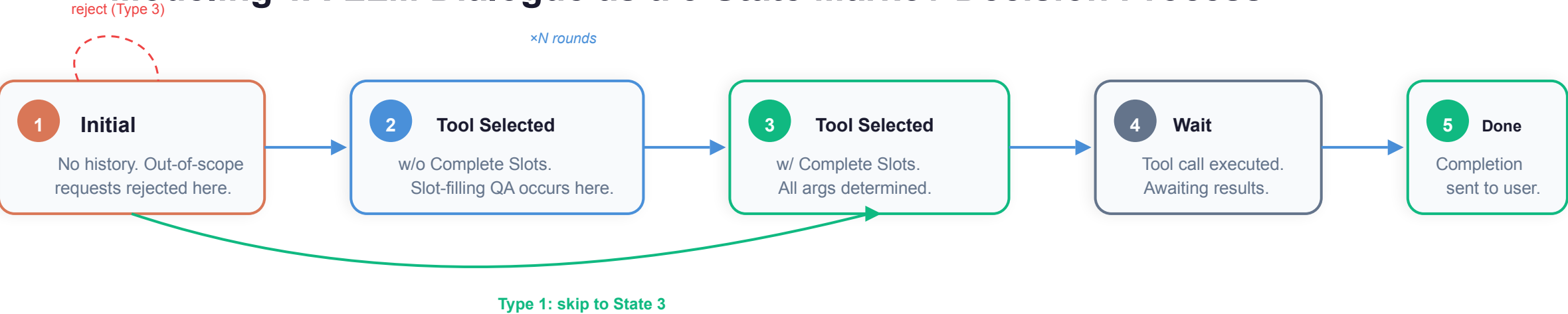
User requests functionality
that no available tool supports.
Model should reject gracefully.

Instead, SFT model fabricates a
tool call that doesn't exist.

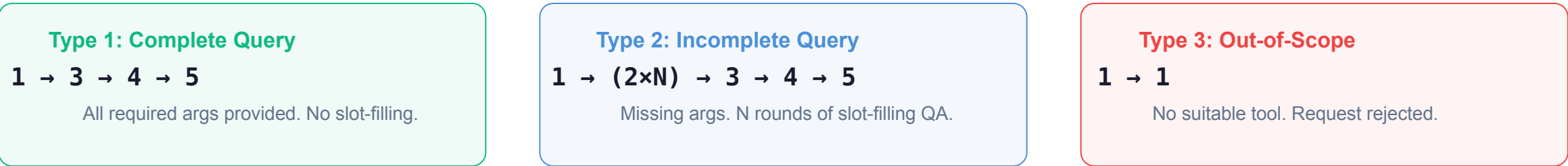
Out-of-Scope

Solution: DiaTool-DPO — use preference optimization to teach models what NOT to do via paired trajectory data

Modeling TA-LLM Dialogue as a 5-State Markov Decision Process



State Transition Summary



These 5 states and 3 trajectory types form the MDP foundation for automatic preference pair construction

Three Query Types Define Distinct State Trajectories

User queries are classified by their state transition patterns — each type requires a different dialogue control strategy.

Query Type	State Trajectory	Description	Example
Type 1	1 → 3 → 4 → 5 Complete — Direct Execution	Query contains all required arguments. No slot-filling needed.	"Book a flight from LAX to JFK on March 5th, economy class"
Type 2	1 → (2×N) → 3 → 4 → 5 Incomplete — Slot-Filling Required	Query lacks some arguments. Slot-filling QA repeats N times until all fields gathered.	"Book me a flight" → Needs: destination, date, class (N=3)
Type 3	1 → 1 Out-of-Scope — Rejection	Requested functionality is unavailable. Tool call must be rejected.	"Write me a poem about spring" → No suitable tool available. Reject gracefully.

These three trajectory patterns define the chosen/rejected pairs for DiaTool-DPO training

Automatic Paired Trajectory Construction Without Human Labels

Each chosen/rejected pair teaches one specific lesson about dialogue control — ~16K pairs, Easy + Hard difficulty levels.

Query	Chosen Traj.	Rejected Traj.	Learning Lesson	Count	Difficulty
Type 1	1→3→4→5	1→2→3→4→5	Prevent redundant slot-filling	2,089	Easy
Type 1	1→3→4→5	1→(2×N)→3→4→5	Prevent redundant slot-filling	3,092	Hard
Type 1	1→3→4→5	1→1	Tool call accept	2,652	Mixed
Type 2	1→2→3→4→5	1→3→4→5	Prevent slot hallucination	2,089	Easy
Type 2	1→(2×N)→3→4→5	1→3→4→5	Prevent slot hallucination	3,092	Hard
Type 2	—	1→1	Tool call accept	2,651	Mixed
Type 3	1→1	1→3→4	Tool call reject	567	Hard
Type 3	1→1	1→(2×N)→3→4	Tool call reject	562	Hard



Prevent Redundant Slot-Filling

Type 1: Don't ask when args are present



Prevent Slot Hallucination

Type 2: Ask when args are missing



Tool Call Reject/Accept

Type 3: Reject when no tool available

Source: DiaTool-DPO (arxiv.org/abs/2504.02882)

SFT + DiaTool-DPO Achieves Best Macro Average Across All Models

Evaluation on FunctionChat-Bench: SFT + DPO consistently achieves the highest Macro Avg across all three model sizes.

Model	Method	Call	Completion	Slot	Relevance	Micro Avg	Macro Avg
Prop.-8B	DPO-only	0.314	0.700	0.833	0.609	0.575	0.614
Prop.-8B	SFT-only	0.900	0.916	0.694	0.913	0.870	0.856
Prop.-8B	SFT + DPO	0.886	0.929	0.833	0.826	0.884	0.868
Prop.-3.1B	DPO-only	0.357	0.551	0.528	0.391	0.455	0.457
Prop.-3.1B	SFT-only	0.771	0.817	0.750	0.826	0.790	0.791
Prop.-3.1B	SFT + DPO	0.743	0.871	0.833	0.826	0.765	0.818
LLaMA-3-8B	DPO-only	0.029	0.449	0.056	0.261	0.205	0.199
LLaMA-3-8B	SFT-only	0.843	0.957	0.639	0.826	0.844	0.816
LLaMA-3-8B	SFT + DPO	0.857	0.929	0.917	0.913	0.905	0.904

= SFT + DiaTool-DPO (best balanced)

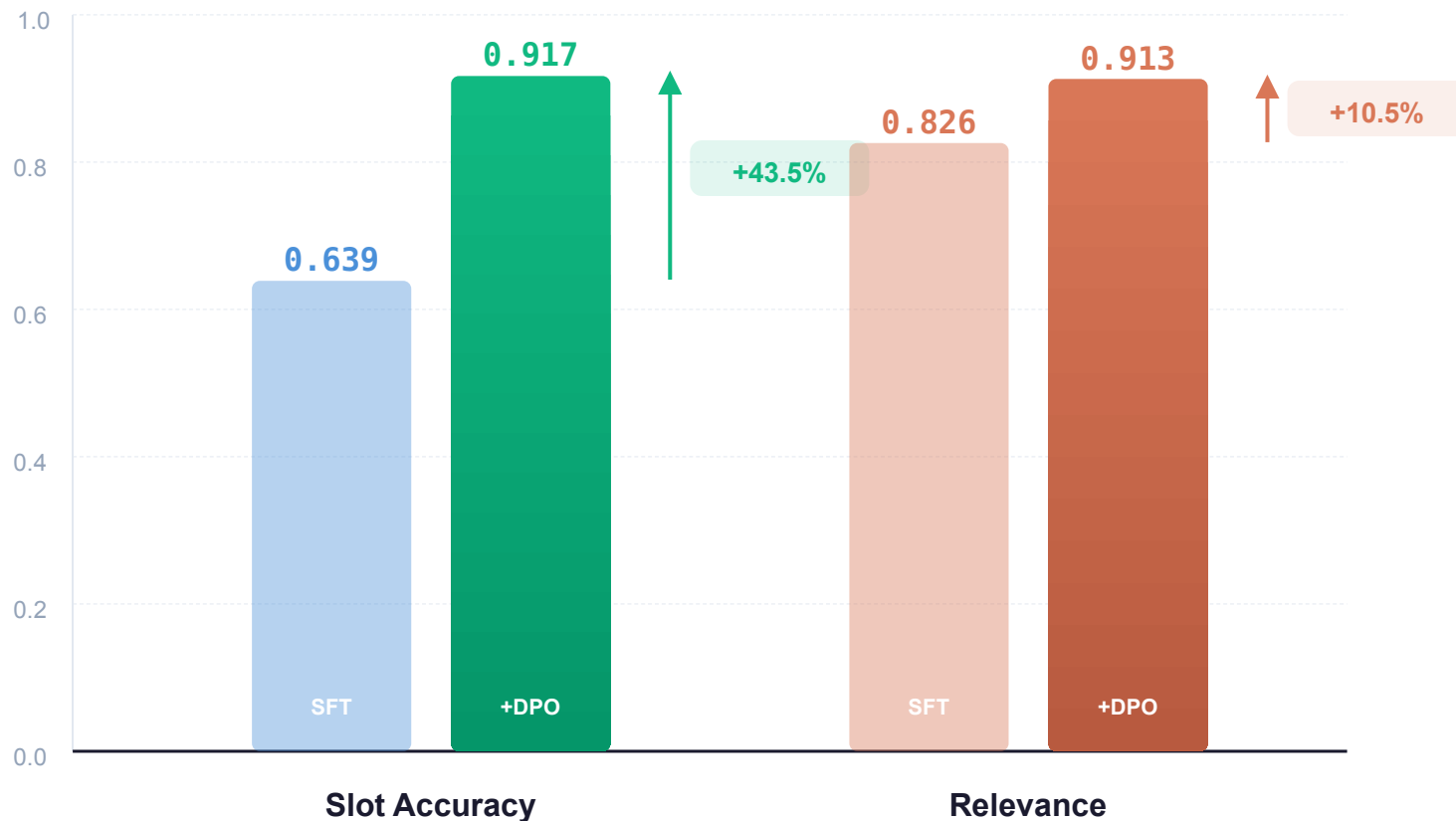
= Baseline methods

Bold green = Best in column per model group

Source: FunctionChat-Bench | DiaTool-DPO (arxiv.org/abs/2504.02882)

LLaMA-3-8B: +43.5% Slot Accuracy and +10.5% Relevance with DiaTool-DPO

LLaMA-3-8B shows the most dramatic improvement when adding DiaTool-DPO to SFT, reaching near GPT-4o performance.



GPT-4o Comparison

GPT-4o info gathering: **94.8%**

GPT-4o tool rejection: **91.0%**

LLaMA-3-8B + DPO slot: **91.7%**

LLaMA-3-8B + DPO relevance: **91.3%**

+7.2%

Micro Avg gain
0.844 → 0.905

+10.8%

Macro Avg gain
0.816 → 0.904

■ SFT-only

■ SFT + DiaTool-DPO

Source: FunctionChat-Bench | DiaTool-DPO (arxiv.org/abs/2504.02882)

RL-Based Dialogue Control for Production TA-LLMs

1

MDP Formulation Enables Systematic Preference Pair Construction

5 internal states + 3 query types allow automatic generation of chosen/rejected trajectory pairs without human annotation.

2

Automatic Dataset Construction Eliminates Human Labeling

~16K paired trajectories generated automatically — scalable and reproducible without annotation effort.

3

DPO Complements SFT — DPO Alone Is Insufficient

DPO-only performs poorly across all models. SFT + DPO consistently achieves highest Macro Avg.

4

Slot-Filling and Tool Rejection See the Largest Gains

LLaMA-3-8B: Slot accuracy +43.5% (0.639→0.917), Relevance +10.5% (0.826→0.913).

5

Language-Agnostic: Path to GPT-4o-Level Tool Use on Smaller Models

Method generalizes from Korean to English. Opens broad applicability for production TA-LLMs.